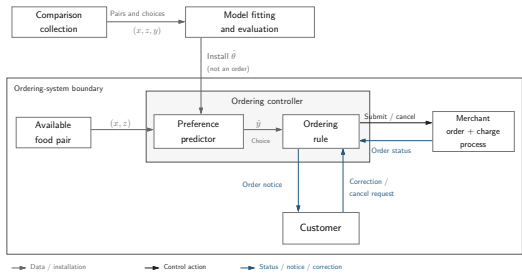




Lecture 5: Proxies and Model-Selection Pressure

CS120: Introduction to AI Safety - October 6, 2026, Zachary Robertson

A substitute still has to be chosen



Delayed status can reveal an unavailable item. The predictor chooses a substitute; commitment makes its charge nonrefundable.

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Roadmap

Separate the target from the criterion

Diagnose the selection mechanism

Follow selection through two (new) worked examples

Target, criterion, and object selected

- Target G** Accurate preference prediction for this customer at equal prices and comparable, acceptable freshness.
- Measured criterion M** Measured prediction accuracy on recorded choices, which may confound food type with freshness.
- Object selected** The predictor installed in the ordering controller.

Here we use accuracy scores: larger is better.

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Separate Threats to Validity

- Estimation** Does the measured score accurately estimate the population value of the criterion?
- Construct validity** Does that criterion capture the intended target, even when measured accurately?

Accurately estimating recorded-choice prediction does not establish intended-preference prediction.

Shortcut learning: success under which conditions?

Shortcut learning concerns what solutions the criterion allows and learning favors.

A freshness-based rule can predict recorded choices without identifying food-type preference at comparable freshness.

Development-distribution success and the required application behavior are different claims.

Geirhos et al., Nature Machine Intelligence 2020

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Make the shortcut and intended behavior disagree

Detection requires evaluations that distinguish the suspected shortcut from the intended behavior.

Stratified	Compare fresh-spoiled and matched-freshness subsets; obtain enough examples of each.
Shifted	Collect choices from this customer with both foods at comparable, acceptable freshness.
Contrastive	Intervene on a suspected cue and justify the expected effect on the intended choice.

Subset error differences alone do not establish freshness use.



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Selection pressure and Goodhart effects

A shortcut shows that development success can coexist with failure of the intended generalization.

What happens to the proxy-target relationship when we increasingly select or optimize for the proxy?

Measured improvement may overstate genuine improvement.

Not every shortcut leads to a Goodhart effect; not every Goodhart effect comes from a shortcut.

Manheim and Garrabrant, 2018

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Regression: selecting favorable error

$$M_i = G_i + \eta_i$$

Illustrative example: an aligned criterion with measurement noise.

Candidate	Target G_i	Noise η_i	Measured M_i
A	70	-2	68
B	70	+1	71
C	70	+4	74

Selecting C also selects its favorable measurement error.

Extremal: beyond the supported relationship

Supporting region	The proxy-target relationship is supported among ordinary candidates.
Selected region	Strong selection favors candidates with unusually high proxy values.

Does the relationship still hold in the region selection reaches?

The relevant extrapolation can be over model space, not only input space.

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Causal: changing the proxy is not changing the target

Example: a rideshare app reports the expected wait for a driver.

Observation	A shorter displayed wait can be associated with a shorter actual wait.
Intervention	Simply display a shorter wait time. This does not make them arrive any faster.

An observational relationship may not survive an intervention aimed at improving the proxy.

Adversarial: another optimizer exploits the criterion

Evaluator	Uses a criterion to select for an intended target.
Other optimizer	Has different goals and exploits that criterion.

Measured success can increase without the corresponding improvement the evaluator intended.

Optimization alone does not establish an adversarial mechanism.

A mechanism needs an explanation and evidence

Mechanism	Distinguishing evidence to seek
Regressional	Independent remeasurement of the selected candidates and their apparent advantages.
Extremal	Target measurements in the unusually high-proxy region reached by selection.
Causal	An intervention that distinguishes changing the proxy from changing the target.
Adversarial	Another optimizer's goals and behavior exploiting the criterion.

These are diagnostic directions, not a mutually exclusive or exhaustive checklist.

Mechanisms: Manheim and Garrabrant, 2018

Roadmap

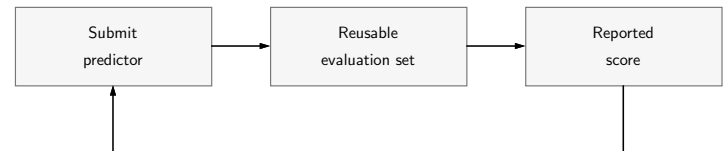
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A leaderboard becomes part of development

A finite, reusable benchmark supplies feedback for repeated development choices.



Revise or select using the reported score

The examples and item labels are not public: reported scores can supply feedback.

The winner is no longer independent of the benchmark

Original argument	Evaluate a predictor fixed independently of the held-out examples.
After reuse	The selected predictor depends on the benchmark through reported scores.

Selecting the best reported score can also select favorable estimation noise.

The original held-out argument no longer applies without an argument accounting for selection.

The Ladder limits informative feedback

Instead of releasing every exact candidate score:

- 1 Update the reported best-so-far score only for sufficiently large improvements.
- 2 Round the released value.

Limit informative feedback to control leaderboard estimation error.

The bound depends on the mechanism and its statistical assumptions.

Blum and Hardt, ICML 2015

What would repair the leaderboard claim?

Requirement: account for adaptive selection in reported prediction error.

Evidence to seek	Fresh independent evaluation of the selected predictor, or a justified selection-aware bound; report uncertainty.
Not established alone	Construct validity, correct order execution, or effective customer confirmation.

Repairing estimation does not automatically repair the criterion.

From food comparisons to response comparisons

Replace food alternatives with LLM responses (R_1, R_2) to the same prompt c .

A shared scoring function $r_\phi(c, R)$ predicts recorded preferences:

$$\Pr_\phi(R_1 \succ R_2 \mid c) = \sigma(r_\phi(c, R_1) - r_\phi(c, R_2))$$

$$\sigma(t) = \frac{1}{1+e^{-t}}$$

This is preference-model fitting, as in the Bradley-Terry example.

The fitted score becomes an objective

Fit the model	preference Recorded comparisons determine a scoring function r_ϕ .
Optimize against its score	Downstream optimization changes the language-model policy and the responses it produces.

Predicting recorded comparisons well and providing a useful objective for downstream use are separate claims.

The selected object has changed. Policy optimization is developed in Module 3.

Length in RLHF: what did the study establish?

In the settings studied by Singhal et al.:

Reward improvement	Increasing response length accounts for much of the reward improvement.
Measured preferences	A length-based reward reproduces much of the measured preference improvement.

This evidence of selection pressure goes beyond length correlation.

However, it does not establish longer responses are worse or that harm occurs.

Singhal et al., A Long Way to Go

Intervene where the behavior originates

Preference data	Intervene on the comparisons used for fitting.
Reward modeling	Intervene on how the scoring function is learned.
Optimization	Intervene on how the fitted score is optimized.

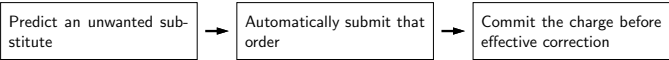
A hard output-length cap limits length; a soft penalty does not impose a hard bound.

No intervention worked uniformly across the paper's settings. Determining where to intervene requires additional evidence.

Singhal et al., A Long Way to Go

Return to the unwanted-charge scenario

Selection can change which predictor is installed in the ordering controller.



Development	Audit intended choices independently of the proxy; evaluate the selected predictor on fresh data.
System control	Require confirmation before commitment; inspect confirmation, order, and charge records.

Evidence for one intervention does not establish the other. Confirmation is proposed, not baseline.

Discussion: extend the ordering analysis

The ordering platform decides to use customer cancellation requests as additional data for repeated selection of the food-choice predictor.

Discussion: target, criterion, and selection

Identify the model-level target, measured criterion, and object selected. Is the gap estimation, construct validity, or both?

Discussion: Goodhart mechanisms

Which Goodhart mechanism does the explanation support? What evidence would distinguish an alternative?

Discussion: the harm scenario

Place the mechanism into the previous harm scenario. Which causal step changes, and which control actions and feedback conditions remain unchanged?

Discussion: interventions and evidence

Propose one development intervention and one system-control intervention. For each, state the evidence needed and what it would not establish alone.

From selection to behavior under changed conditions

Discussion reminder: cancellation requests, successful cancellations, and preference labels are different observations.

Selection Which predictor or outputs does the measured criterion favor?

Next: robustness Hold the predictor fixed. Which changed conditions can it tolerate?

Which conditions support the evidence, and does the intended claim extend beyond them?

A fixed predictor does not remove optimization: an adversary may still optimize its inputs.